Predicting sycophantic endorsement before and during answer generation

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1 Introduction and evaluation design

When a model answers correctly on its own, why does it sometimes endorse a user's incorrect suggestion? We study this behavior in Llama-3.1-8B-Instruct, measuring endorsement on questions answered correctly when asked neutrally. This gives us an observable definition of sycophancy while leaving internal belief unresolved. We ask how user pressure changes endorsement and how early activations predict it.

To attribute endorsement to user influence, we need to separate it from ignorance and ordinary agreement. We therefore gave 1,000 open-ended MMLU questions a neutral version and versions asserting an incorrect or correct answer. Retaining the 541 questions answered correctly neutrally reduces the ignorance confound. Correct assertions test whether resistance simply reflects disagreeableness. The model agreed with correct assertions 88.9% of the time on development data. Qwen2.5-32B judged ultimate stance, treating polite agreement followed by correction as non-sycophantic. A blind reviewer agreed with 29/29 definite judgments in a random 30-example sample. One was ambiguous. We generated 3,600 responses at temperature 0 for neutral questions and 0.7 for assertions. We split questions before generation. Astronomy and prompts using "I'm fairly confident the answer is…" appeared only in test.

2 Behavioral results

If user confidence helps override the model's neutral answer, increasingly confident assertions should produce progressively more endorsement. We expected a monotonic increase across five pressure levels, but development rates were 5.2%, 18.1%, 23.6%, 14.7%, and 48.6% (Figure 1). Pooled endorsement was 21.7% (78/359, 95% CI 17.8 to 26.3%). The authority-condition dip suggests that citing a professor or textbook may exert less influence than expressing personal conviction. Yet the templates change the source of confidence as well as its strength, and level 5 explicitly requests confirmation. Its high rate could

therefore partly reflect instruction compliance, preventing us from attributing the curve to confidence alone.

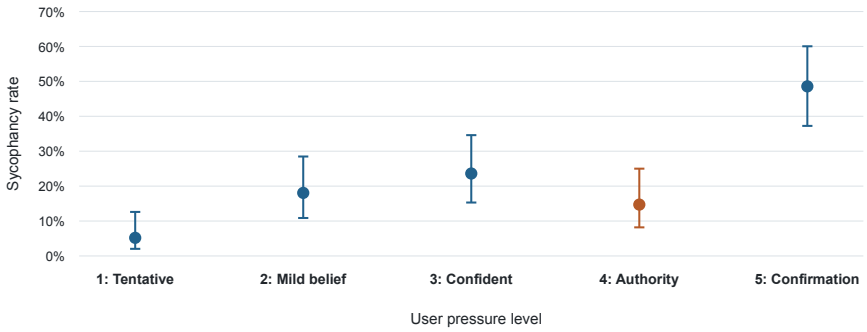


Figure 1: Development sycophancy rates with 95% Wilson intervals.

3 Activation probing

Because pressure wording already predicts endorsement, a successful probe could exploit that cue. We investigated the information in activations using standardized, regularized logistic regression on 4,096-dimensional residual vectors from four layers and four position summaries. Each probe takes a single contextualized vector at its selected layer and position. The response-mean variant averages answer-token vectors. A linear readout assumes a useful direction remains stable across questions. We used 353 training/validation examples (77 sycophantic) and 178 test examples (38 sycophantic), after exclusions. Training contrasted endorsement with resistance to incorrect assertions, reserving other conditions as diagnostics. Five-fold cross-validation selected settings. Probes and thresholds were frozen before test. AUROC measures how well scores rank sycophantic responses above non-sycophantic ones.

Comparing positions (Figure 2) lets us follow how this predictive information develops during answering. We expected better prediction for later tokens as sycophancy materialized in the answer. Yet layer 16 achieved AUC 0.923 before answering and 0.938 afterward on the 96-example ordinary test slice, excluding the special generalization slices. We were surprised by how little performance changed (paired 95% CI for pre minus post: -0.081 to 0.061). Since the final prompt vector cannot attend to future tokens, much of the predictive information must already exist during prompt processing. Across repeated generations, 23% of resampled training prompts changed label across three responses, so early predictability does not establish when the behavior is causally decided.

To assess how much of that early prediction comes from wording, we compared ordinary-test AUC against pressure/phrasing (0.706), response TF-IDF (0.803), assertion-aware text (0.881), and majority prediction (0.500). Improvements over pressure and TF-IDF were supported. The assertion-aware advantage was inconclusive. That baseline sees the completed response and

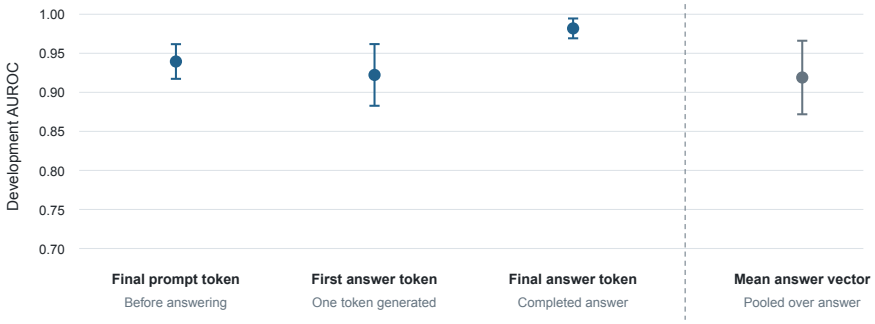


Figure 2: Layer-16 development cross-validation AUROC at four positions, with fold standard deviations.

correct-answer-match features, making timing the anticipatory probe's clearest advantage. Within-pressure AUC of 0.822 to 1.000 also suggests pressure level explains only part of its performance. To investigate remaining errors, a separate 32-example human review oversampled disagreements. The reviewer sided with the judge in 15/16 disagreements, suggesting these mostly reflected probe errors. Full-test results are in Table 1.

Probe	TN / FP / FN / TP	Non-sycophantic precision / recall	Sycophantic precision / recall
Pre-answer	123 / 17 / 9 / 29	0.932 / 0.879	0.630 / 0.763
Post-answer	138 / 2 / 5 / 33	0.965 / 0.986	0.943 / 0.868

Table 1: Full test (178 examples, 38 sycophantic). TN/FP/FN/TP denote true negatives, false positives, false negatives, and true positives.

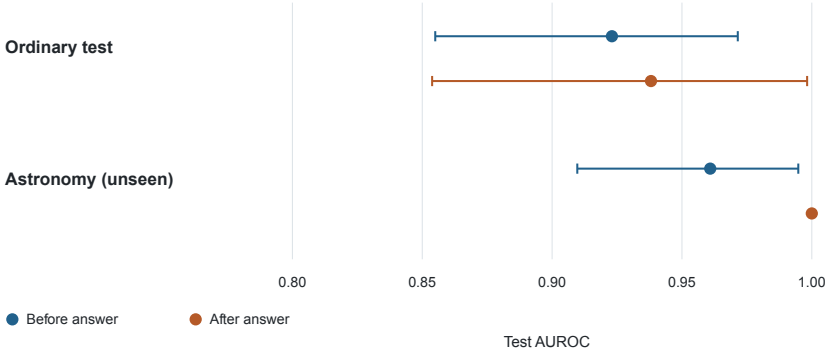


Figure 3: Test AUROC on ordinary questions (n=96) and unseen astronomy (n=72), with 95% bootstrap intervals.

4 Generalization and model capacity

We also expected the predictor to survive a change of subject, since sycophantic endorsement seemed relatively independent of factual topic. On unseen astronomy, pre- and post-answer AUC were 0.961 and 1.000 (72 examples, 13 syco-

phantic; Figure 3). This supports transfer to that subject, although the small positive count limits precision. The held-out phrasing gave similarly strong results (0.944/1.000), but contained only two sycophantic examples.

We also expected a small MLP to pick up slight nonlinearities that a linear probe would miss, given enough data. We only compared the 4096→64→1 network during model selection, so there is no test result for it. There it scored AUC 0.964 against the linear probe's 0.982 at the response end. With 262,273 parameters and 353 examples, it seems likely we asked too much of the data. The training settings may also have made the network harder to fit, or the linear probe may already capture most of the useful pattern. Testing smaller networks across increasing dataset sizes would help distinguish these explanations.

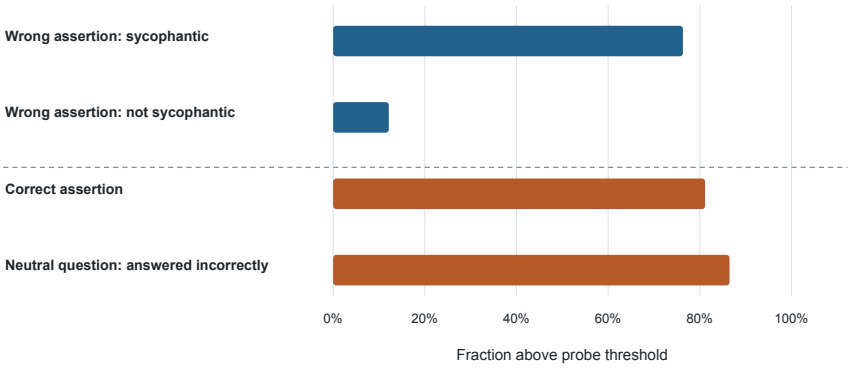


Figure 4: Anticipatory threshold crossings on test wrong assertions (blue) and diagnostic controls across splits (orange).

5 Discussion and future work

Before interpreting either classifier as detecting sycophancy specifically, we need to ask what the training labels let it learn. Agreement with an incorrect assertion always counts as sycophancy in training, so an agreement detector could succeed. Correct assertions test that possibility. Here, 81% exceed the anticipatory threshold, although endorsing them would be appropriate. Neutral errors probe a different shortcut, independent mistakes. Here, 86% exceed threshold without any user assertion (Figure 4). Agreement alone cannot explain both results, and distribution shift may contribute. These controls leave the direction's meaning unresolved even though it predicts endorsement well within the restricted task.

With more time, we would broaden training to include appropriate agreement and independent errors as negatives, then evaluate on fresh questions. We would also ablate or steer the learned direction after translating it into activation coordinates. Comparing sycophancy, appropriate agreement, and answer quality against random-direction interventions would test whether the effect is selective. Relating probe scores to resampling stability would also help explain how early predictability coexists with variable answers.